

1 Part 13: Brownian Motion Case Study

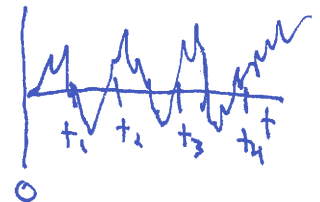
2 Summary

- 3 This Part presents a simple, but important, example of how careful deriva-
4 tion of the MLE and its distribution can lead to useful insight.
- 5 We begin by defining some standard concepts regarding Brownian motion.

1 *The Standard Brownian Motion (Wiener Process)*

- 2 A stochastic process $\{W(t) : t \geq 0\}$ is a (standard) Brownian motion if the
3 following hold:

- 4 1. $W(0) = 0$. (almost surely)
- 5 2. $W(t)$ is continuous as a function of t . (almost surely)
- 6 3. $W(t_4) - W(t_3)$ and $W(t_2) - W(t_1)$ are independent for $t_1 \leq t_2 \leq t_3 \leq t_4$.
- 7 4. The distribution of $W(t_2) - W(t_1)$ depends only on $|t_1 - t_2|$ for t_1, t_2 .
- 8 5. $W(t)$ is $\text{Normal}(0, t)$ for all t .



1 *Brownian Motion with Drift and Scaling*

2 A stochastic process $\{B(t) : t \geq 0\}$ is a **Brownian Motion with drift μ and**
 3 **scaling σ^2** if $\{W(t) : t \geq 0\}$ is a standard Brownian motion and

$$4 \quad B(t) = \underbrace{\mu t}_{\text{drift}} + \sigma W(t).$$

5 Note that this implies the following:

- 6 1. $B(0) = 0$. (almost surely)
- 7 2. $B(t)$ is continuous as a function of t . (almost surely)
- 8 3. $B(t_4) - B(t_3)$ and $B(t_2) - B(t_1)$ are independent for $t_1 \leq t_2 \leq t_3 \leq t_4$.
- 9 4. The distribution of $B(t_2) - B(t_1)$ depends only on $|t_1 - t_2|$ for t_1, t_2 .
- 10 5. $B(t)$ is Normal($\mu t, \sigma^2 t$) for all t .

$$\begin{aligned} E(B(t)) &= E(\mu t + \sigma W(t)) \\ &= E(\mu t) = \mu t \end{aligned}$$

1 *Geometric Brownian Motion*

2 A stochastic process $\{S(t) : t \geq 0\}$ is a **geometric Brownian motion** if $\{B(t) :$
 3 $t \geq 0\}$ is a Brownian motion with drift μ and scaling σ^2 and

$$4 \quad S(t) = S(0) e^{B(t)}$$

5 where $S(0) > 0$ is the initial value. Define $\alpha = \mu + \sigma^2/2$.

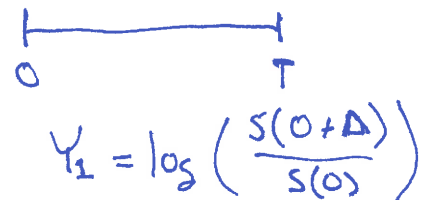
6 Note that this implies the following:

- 7 1. $S(t)$ is continuous as a function of t . (almost surely)
- 8 2. $S(t_4)/S(t_3)$ and $S(t_2)/S(t_1)$ are independent for any $t_1 \leq t_2 \leq t_3 \leq t_4$.
- 9 3. The distribution of $S(t_2)/S(t_1)$ depends only on $|t_1 - t_2|$ for any t_1, t_2 .
- 10 4. $S(t)$ is Lognormal($\log(S(0)) + \mu t, \sigma^2 t$) for all t .
- 11 5. $S(t)$ has expected value $S(0)e^{\alpha t} = S(0)e^{(\mu + \sigma^2/2)t}$
- 12 6. $\log(S(t_2)/S(t_1))$ is Normal($\mu(t_2 - t_1), \sigma^2(t_2 - t_1)$) for all $t_1 \leq t_2$.

- 1 Let's consider the geometric Brownian motion model. Can we use data
- 2 (observed prices) to estimate α and σ^2 ?
- 3 Suppose we observe the stock price over the period $[0, T]$. We will model
- 4 the price as a geometric Brownian motion $\{S(t) : t \geq 0\}$.
- 5 Divide the time interval into n equal-width subintervals, with $\Delta = T/n$.

- 6 Define, for $i = 1, 2, \dots, n$,

$$Y_i = \log \left(\frac{S(t_{i-1} + \Delta)}{S(t_{i-1})} \right)$$



$$Y_1 = \log \left(\frac{S(0+\Delta)}{S(0)} \right)$$

- 7 where $t_i = (i/n)$ the times at which the price is observed.

- 8 From Property 6 of a geometric Brownian motion, $Y_1, Y_2, \dots, Y_n$ are i.i.d.
- 9 $N(\Delta\mu, \Delta\sigma^2)$.

- 1 **Exercise:** Find the MLE of ~~θ~~ (α, σ^2) . Approximate the distribution of ~~θ~~ ^{this pair.}

The $Y_1, Y_2, \dots, Y_n$ are iid $N(\Delta\mu, \Delta\sigma^2)$

Let $\Theta_1 = \Delta\mu$, $\Theta_2 = \Delta\sigma^2$ "original parameters"

$$\underline{\Theta} = (\Theta_1, \Theta_2)$$

We know from previous derivations that $\hat{\underline{\Theta}}_{MLE}$ is approx. normal with covariance matrix

$$\underline{I}^{-1}(\underline{\Theta}_0)/n = \begin{bmatrix} \Theta_2/n & 0 \\ 0 & 2\Theta_2^2/n \end{bmatrix}$$

But, want covariance matrix for (α, σ^2)

In the notation of the Delta Method,

$$\alpha = g_1(\underline{\theta}) = \mu + \sigma^2/2 = \frac{\theta_1}{\Delta} + \frac{\theta_2}{2\Delta}$$

$$\sigma^2 = g_2(\underline{\theta}) = \frac{\theta_2}{\Delta}$$

By invariance prop. of MLE,

$$\hat{\alpha}_{MLE} = \frac{\hat{\theta}_1}{\Delta} + \frac{\hat{\theta}_2}{2\Delta} \quad \text{where} \quad \hat{\theta}_1 = \bar{Y}$$

$$\hat{\theta}_2 = \left(\frac{n-1}{n}\right) S_Y^2$$

$$\hat{\sigma}_{MLE}^2 = \frac{\hat{\theta}_2}{\Delta}$$

The matrix $\underline{G}$ in this case:

$$\underline{G} = \begin{bmatrix} \frac{\partial g_1(\underline{\theta})}{\partial \theta_1} & \frac{\partial g_1(\underline{\theta})}{\partial \theta_2} \\ \frac{\partial g_2(\underline{\theta})}{\partial \theta_1} & \frac{\partial g_2(\underline{\theta})}{\partial \theta_2} \end{bmatrix}$$

$$= \begin{bmatrix} 1/\Delta & 1/2\Delta \\ 0 & 1/\Delta \end{bmatrix}$$

$$\begin{aligned}
 & S_0, \quad G \left(I^{-1}(\theta_0)/n \right) G^T \\
 & = \begin{bmatrix} \frac{\theta_2}{n\Delta^2} + \frac{\theta_2^2}{2n\Delta^3} & \theta_2^2/n\Delta^2 \\ \theta_2^2/n\Delta^2 & 2\theta_2^2/n\Delta^2 \end{bmatrix} \quad \text{note } T=n\Delta \\
 & \quad \sigma^2 = \theta_2/\Delta \\
 & = \begin{bmatrix} \sigma^2/T + \sigma^4/2n & \sigma^4/n \\ \sigma^4/n & 2\sigma^4/n \end{bmatrix} \quad \text{The covariance matrix for } (\hat{\alpha}, \hat{\sigma}^2)
 \end{aligned}$$

Exercise: What does this result suggest regarding how large n should be chosen for the best estimation of σ^2 ? What about for α ?

In order to estimate σ^2 well, choose n arbitrarily large since $V(\hat{\sigma}^2) = 2\sigma^4/n$

But, the estimation of α is limited by T since $V(\hat{\alpha}) = \sigma^2/T + \sigma^4/2n$. Need to increase T in order to drive $V(\hat{\alpha})$ to 0.

Typical values of σ around 0.2. So, if want $SE(\hat{\alpha}) \leq 0.01$, i.e. $\sqrt{\sigma^2/T + \sigma^4/2n} \leq 0.01$
need $T \geq 400$ years

- 1 Is it, in fact, a good idea to sample more frequently in order to estimate
- 2 σ^2 ? See Ait-Sahalia, Mykland, and Zhang (2005), "How Often to Sam-
- 3 ple a Continuous-Time Process in the Presence of Market Microstructure
- 4 Noise," *The Review of Financial Studies*, Volume 18.

Part 14: Model Selection

Summary

- Until now, we have **assumed** a particular **model** for the observable data, i.e., assumed a form for $f(x; \theta)$, known up to parameter θ .
- How do we select from among a list of candidate models? E.g., Normal versus t versus ...
- If we only seek to maximize the likelihood over candidate models, we will be led to **overfitting** to the sample.
- Fortunately, there are formal approaches to **model selection** that can help guide our choices and avoid overfitting.

Example: Alternatives to the Lognormal Pricing Model

- Here, let S_t denote the closing price of some equity on day t .
- If we assume that $S_t = S_{t-1}X_t$, where X_t has the $\text{lognormal}(\mu, \sigma^2)$ distribution, i.e., $\log(X_t)$ has the $\text{Normal}(\mu, \sigma^2)$ distribution. Further assume that $X_1, X_2, \dots$ are independent.
- This is the **lognormal pricing model** that underlies a lot of financial theory, including Black-Scholes.
- One can motivate this model by assuming that $X_t = Y_1 Y_2 \cdots Y_m$, where the Y_i are i.i.d. The CLT states that $\log(X_t)$ will be approximately normal.
- Unfortunately, the model is known to not fit well to real data. In particular, log returns have heavier tails than the normal distribution.

- 1 Previously we considered returns for Kellogg's (K) from 2010 to 2016:

```
> Kellogg = getSymbols("K",  
+   from="2010-1-1",to="2016-12-31", auto.assign=F)  
>  
> ldrK = data.frame(dailyReturn(Ad(Kellogg), type="log"))
```

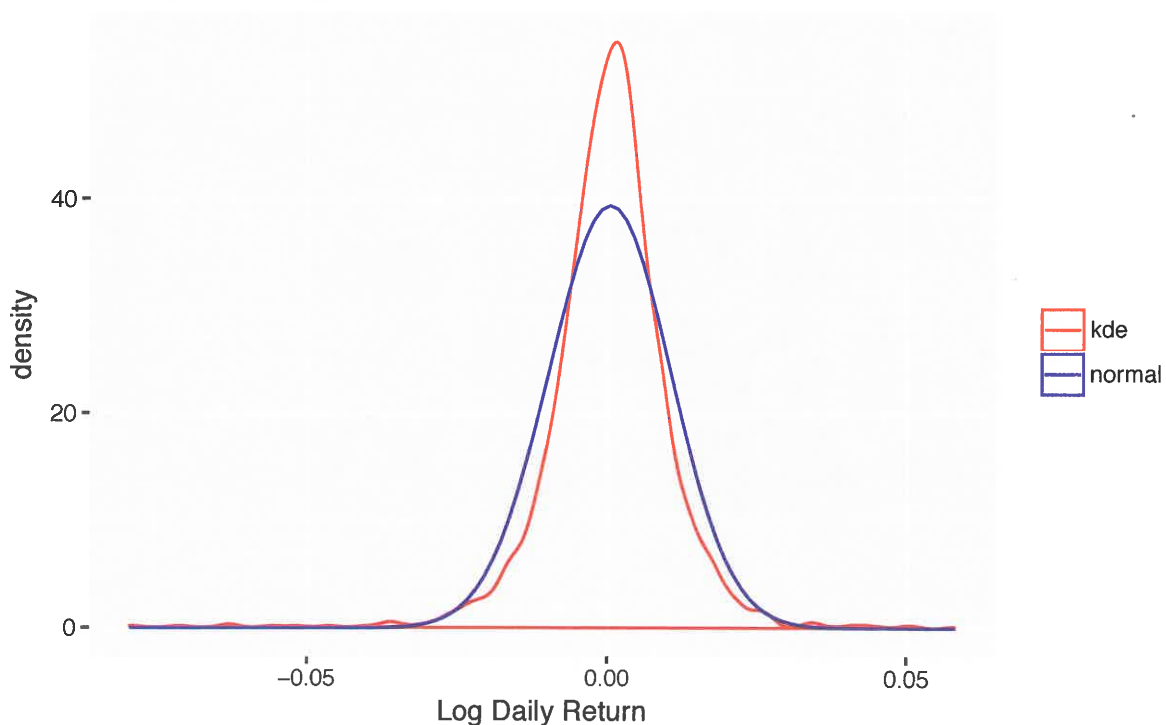
- 2 We constructed the figures on the following slides to show the deficiencies
3 in the normal assumption. They compare the best estimate from a kernel
4 density estimator with the best-fitting normal distribution.

- 5 The second plot shows the densities on the log scale in order to emphasize
6 the discrepancy in the tails.

- 7 The deficiencies in the normal distribution have led people to consider
8 alternatives. Here we will look at a few and then compare their fit to our
9 return data.

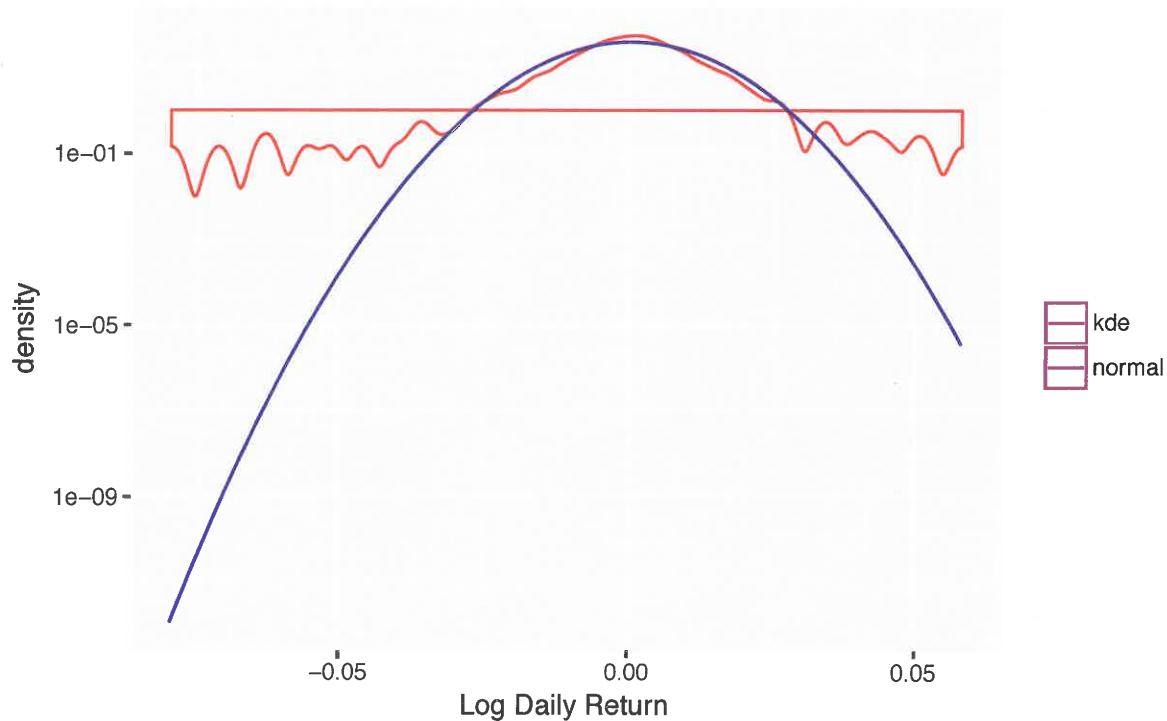
Data for Kellogg's (K)

January 2010 through December 2016



Data for Kellogg's (K)

January 2010 through December 2016



1 The Generalized Error Distribution (GED)

2 Random variable X has the GED with mean μ , variance σ^2 , and shape ν if

$$3 \quad f_X(x) \propto \exp\left(-\frac{1}{2} \left| \left(\frac{X - \mu}{\sigma} \right) \left(\frac{1}{\lambda_\nu} \right) \right|^\nu\right), \quad -\infty < x < \infty, \quad \nu > 0, \quad \sigma^2 > 0$$

4 where

$$5 \quad \lambda_\nu = \left(\frac{2^{-2/\nu} \Gamma(1/\nu)}{\Gamma(3/\nu)} \right)^{1/2}.$$

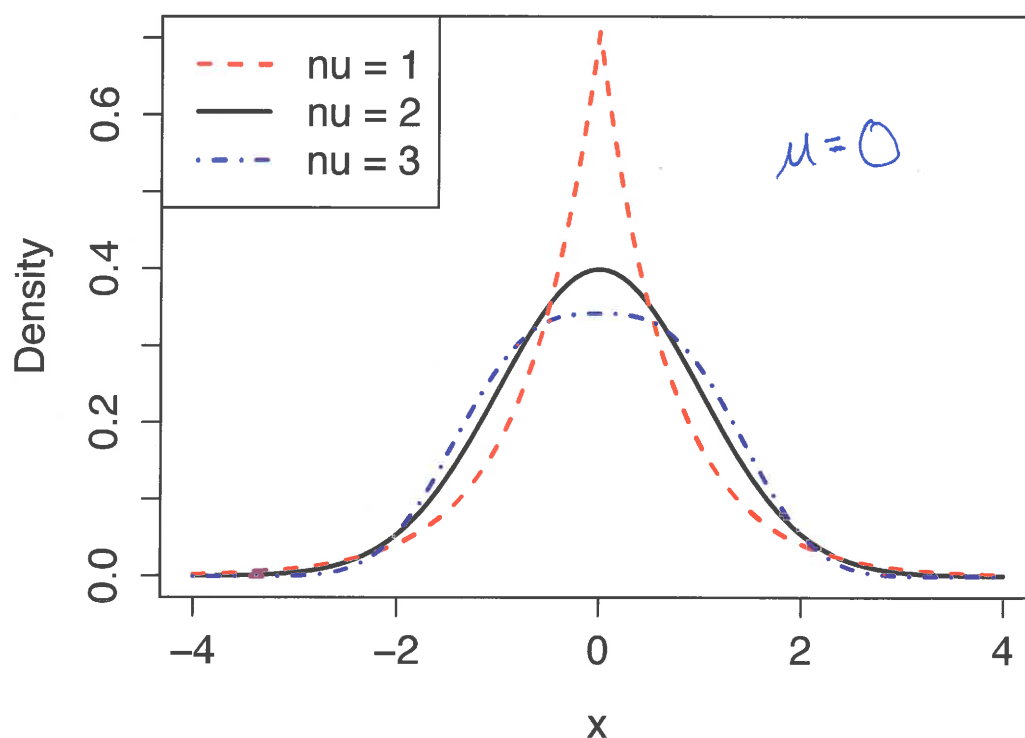
6 Comments:

7 1. When $\nu = 2$, this is the normal(μ, σ^2) distribution. When $\nu = 1$, this is
 8 the **double exponential** or **Laplace** distribution.

9 2. The distributions are all symmetric around μ .

10 3. Package `fGarch` has functions `dged()`, `rged()`, `pged()`, `qged()`.

- 1 The figure below shows the effect of varying ν on the shape of the GED.



2

1 The Nonstandard t -distribution

- 2 Assume T has the (standard) t -distribution with ν degrees of freedom
 3 where $\nu > 2$. Then define

$$X = \mu + \underbrace{\sigma \left(\frac{\nu - 2}{\nu} \right)^{1/2}}_{\text{scale factor}} T$$

$$E(T) = 0 \quad \nu \geq 2$$

$$V(T) = \frac{\nu}{\nu - 2}, \quad \nu > 2$$

- 5 where $\sigma > 0$.

- 6 This distribution will have the “shape” of the t -distribution, but will have
 7 mean μ and variance σ^2 .

- 8 Package `fGarch` has functions `dstd()`, `pstd()`, `qstd()`, `rstd()`.

1 Theodossiou's Skewed, Generalized t Distribution

2 From Theodossiou, "Financial Data and the Skewed Generalized T Distri-
3 bution," *Management Science*, 1998.

4 This distribution always has its mode (peak) at zero, but can be **skewed**
5 (asymmetric).

6 There are four parameters:

7 1. σ^2 , the variance. $\sigma^2 > 0$

8 2. k , which controls the height at the peak. $k > 0$

9 3. n , which controls the probability in the tails. $n > 2$

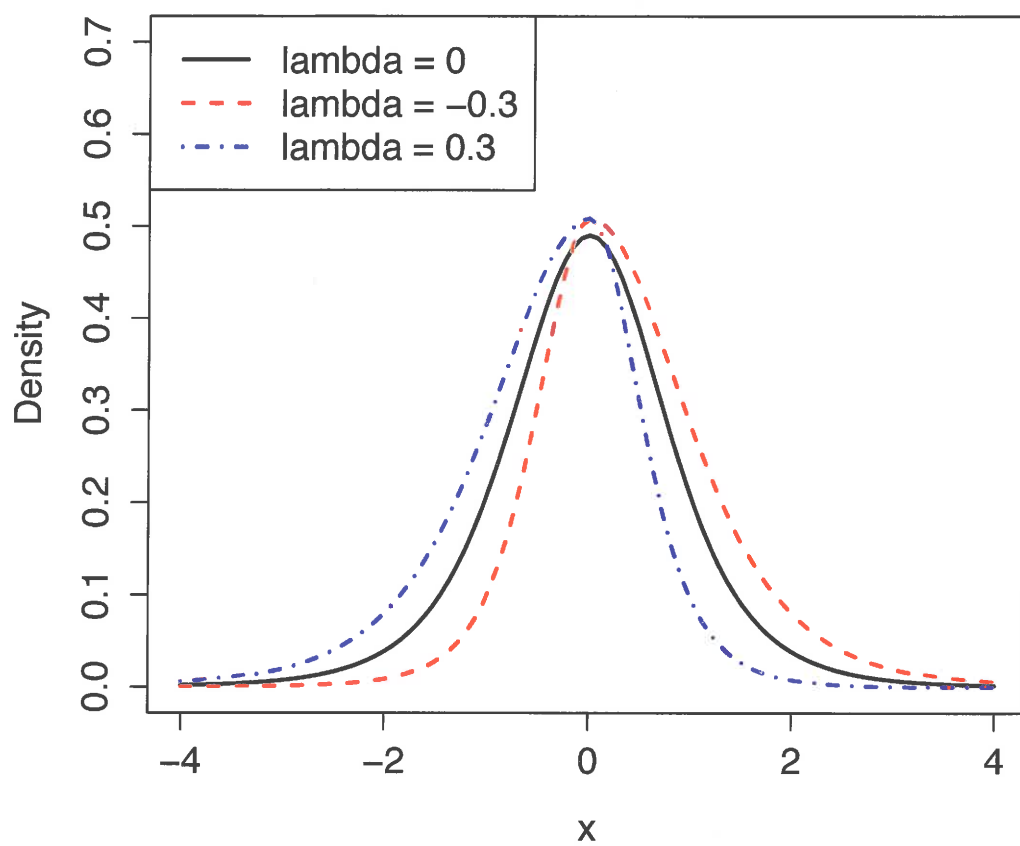
10 4. λ , which controls the amount of skew. $-1 < \lambda < 1$

1 When $\lambda = 0$ and $k = 2$, this matches the nonstandard t distribution with n
2 degrees of freedom, variance σ^2 , and $\mu = 0$.

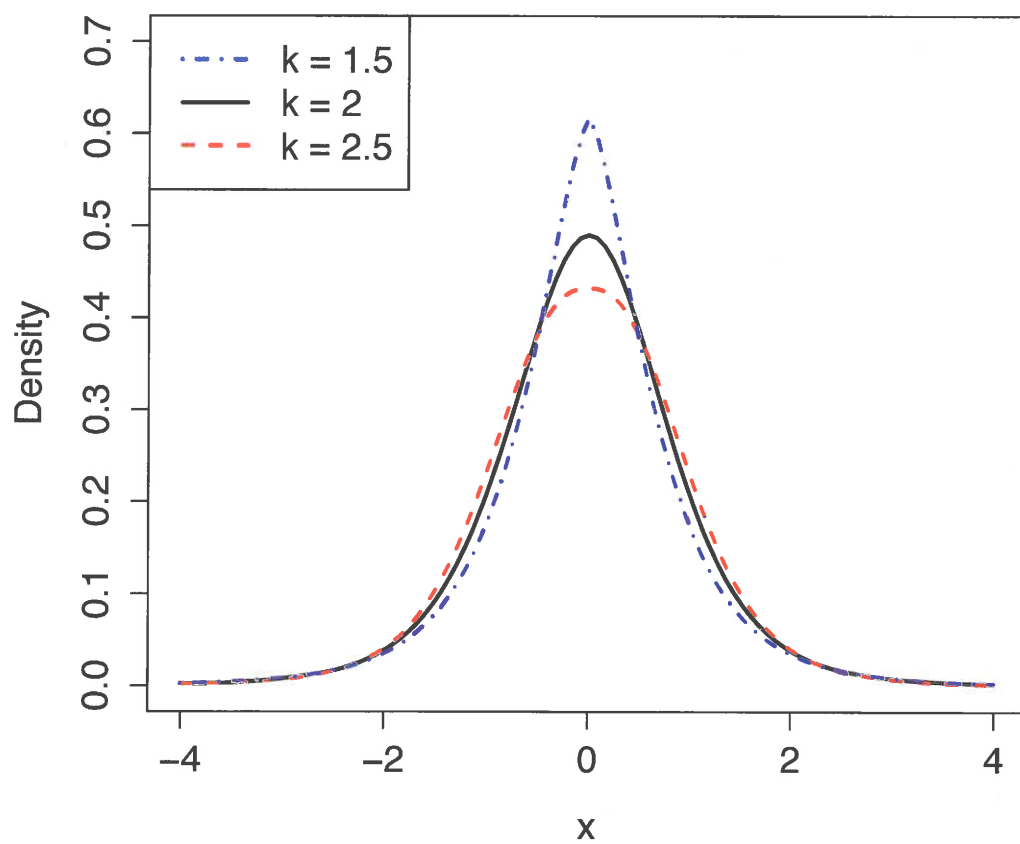
3 Equation (7) in the paper gives the mean of the distribution. As is often the
4 case, more complexity in the distribution leads to more complexity in the
5 expressions for quantities such as the mean.

6 The plots on the following slides give a sense of the effect on the density
7 of varying the parameters λ and k .

8 We could easily generalize this distribution a bit further, and say that X
9 has the **shifted, skewed generalized T distribution** with shift parameter μ
10 if $X - \mu$ has the non-shifted distribution. Note that $E(X) = \mu$ if and only
11 if $\lambda = 0$.



1



1

1 Fitting the Models

- 2 We will use maximum likelihood to fit each of these models to the Kel-
3 logg's data. The following command loads useful functions:

```
> source("http://www.stat.cmu.edu/~cschafer/MSCF/ModelSelectionExample.txt")
```

- 4 The GED and nonstandard t distributions are fit using `FitGED()` and
5 `FitNonStdT()`, respectively:

```
> gedout = FitGED(ldrK$daily.returns)
> print(gedout$mle)
           mu           sigma           nu
0.0005180455 0.0097891288 1.0209490127

> nonstdTout = FitNonStdT(ldrK$daily.returns)
> print(nonstdTout$mle)
           mu           sigma           nu
0.000475831 0.010320269 3.411594515
```

- 1 There is a function `dskewT()` for evaluating the density of the skewed
2 generalized t distribution. Note that the parameters are provided to the
3 function as k , n , λ , σ^2 , and μ (if the shift is included).
- 4 You will note that this function calls `dskewTb()` which utilizes an alterna-
5 tive parameterization. See Equations (26) and (27) in the paper (page 1655)
6 for details.
- 7 This reparameterization is performed in order to make the maximization
8 of the likelihood more stable.

1 We now attempt to find the MLE for this distribution, using the function

2 `FitSkewT()`:

```
> skewtout = FitSkewT(ldrK$daily.returns)
> print(skewtout$mle)
           k           n      lambda      sigma2
2.0109292619 3.4001822258 0.0286725735 0.0001070537
```

3 Next, fit the skewed distribution, with the shift:

```
> skewtshiftout = FitSkewT(ldrK$daily.returns,
+                           allowshift = T)
[1] "Convergence of optim() failed. convergence = 1"
```

4 The exit condition `convergence=1` implies that the argument `maxit` is
 5 not set large enough, i.e., the algorithm is reaching the maximum number
 6 of iterations without finding the minimum to the desired tolerance.

1 Instead, we try the following:

```
> skewtshiftout = FitSkewT(ldrK$daily.returns, allowshift=T,
+                           control=list(maxit=1000))
> print(round(skewtshiftout$mle, 6))
           k           n      lambda      sigma2      mu
1.935187   3.538754 -0.024888  0.000105  0.000724
```

2 We have fit all of our candidate models except for the normal:

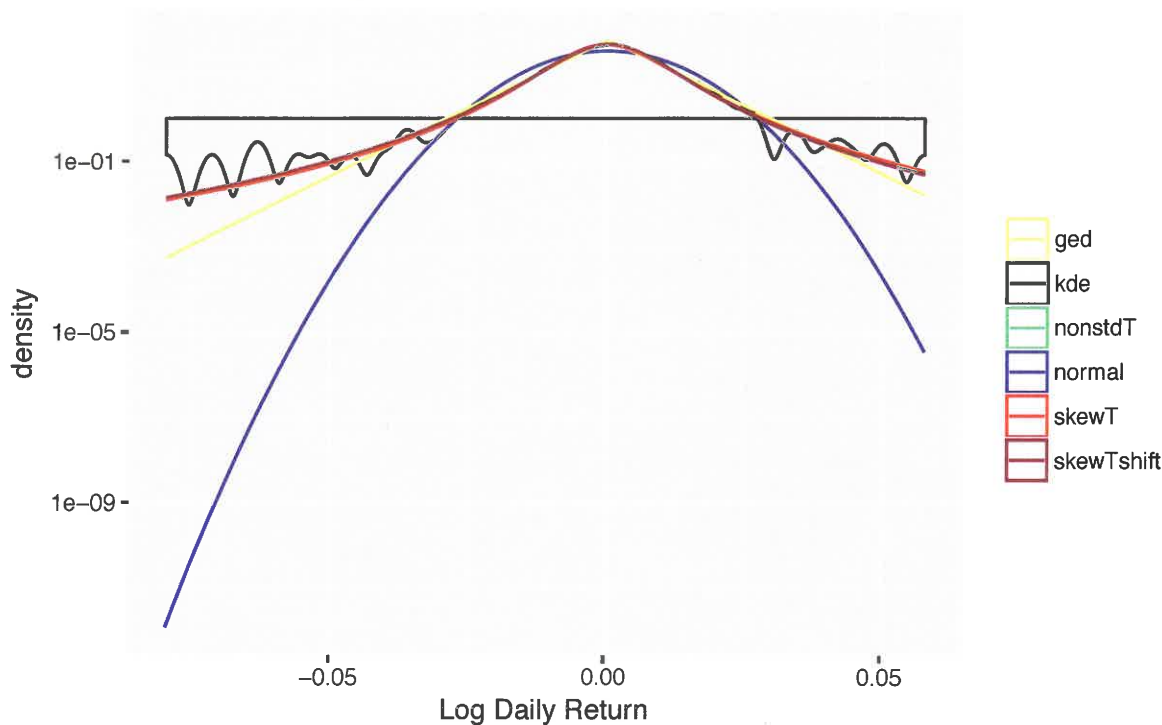
```
> n = length(ldrK$daily.returns)
> normalout = c(mean(ldrK$daily.returns),
+               sqrt(var(ldrK$daily.returns) * (n-1)/n))
> print(normalout)
[1] 0.0003087823 0.0101207134
```

$\hat{\mu}$

$\hat{\sigma}$

Data for Kellogg's (K)

January 2010 through December 2016



- Compare all of these models by calculating the log likelihood for each:

```
> print(sum(dged(ldrK$daily.returns,gedout$mle[1],gedout$mle[2],
+               gedout$mle[3],log=T)))
[1] 5770.239

> print(sum(dstd(ldrK$daily.returns,nonstdTout$mle[1],
+               nonstdTout$mle[2],nonstdTout$mle[3],log=T)))
[1] 5790.015

> print(sum(dSkewT(ldrK$daily.returns,skewtout$mle[1],
+               skewtout$mle[2], skewtout$mle[3],skewtout$mle[4],log=T)))
[1] 5788.42

> print(sum(dSkewT(ldrK$daily.returns,skewtshiftout$mle[1],
+               skewtshiftout$mle[2],skewtshiftout$mle[3],
+               skewtshiftout$mle[4],skewtshiftout$mle[5],log=T)))
[1] 5790.34

> print(sum(dnorm(ldrK$daily.returns,normalout[1],normalout[2],log=T)))
[1] 5592.998
```